

Phase D5 Final Closure: Sixty-Test Operational Library Established Across Six Categories, Ten Prior-Art Claims, Combined Series D Scope of 191 Notes Through Phase D5, With Transition to Phase D6 Boundary Cases

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

This note is D5.15, the fifteenth and final note in Phase D5 of the Series D derivation program. Phase D5 established the operational verification methodology for Paper 3's Full Aspect Integration (FAI) governance architecture—the “how to verify” counterpart to Phase D2's “what governance must do.” Phase D5 produced a sixty-test operational library organized across six categories, ten prior-art claims demonstrating that Paper 3's governance commitments are binary-testable and implementable by non-specialist practitioners, and supporting infrastructure including a consolidated reference, calibration guidance, scenario-specific subsets, a validation framework, and health monitoring integration. With Phase D5 complete, the combined Series D scope through note #650 stands at 191 notes spanning Phase D0 through Phase D5, covering governance requirements, failure modes, multi-commitment interactions, and verifiability. This note formally closes Phase D5 and introduces Phase D6, which adds boundary case precision—explicit establishment of where the architecture's scope boundaries lie—as the final prior-art precision layer before Series D's completion.

1. The Relationship Between Phase D2 and Phase D5

Phase D2 (notes #496–#575, 80 notes) and Phase D5 (notes #636–#650, 15 notes) address different questions that together constitute a complete prior-art treatment of Paper 3's FAI governance architecture.

Phase D2 established what FAI governance must do. Working from Paper 3's six claims, Phase D2 developed the operational variants, sub-commitments, and decompositions of each governance requirement—the shared substrate construction, the exchange boundary, the conflict-handling tier, the evolution feed routing, the configurable dimensions enumeration, and the population-scale dynamics. A practitioner reading Phase D2 knows what an FAI event is required to do and what properties its governance must carry. Phase D2 is a specification: it names the requirements.

Phase D5 established how to verify that governance did what it must. Working from Phase D2's requirements and the 20-test baseline established at D2.24, Phase D5 expanded the test library to 60 tests across six categories, each test binary-decidable (pass/fail), each test derivable without specialist expertise, and each test traceable to a specific Paper 3 commitment. A practitioner reading Phase D5 can take any claimed FAI governance implementation and determine, without ambiguity, whether it satisfies or fails each requirement. Phase D5 is a verification methodology: it provides the testing apparatus the specification requires.

Neither phase replaces the other. Phase D2 without Phase D5 produces requirements whose satisfaction remains contestable—an implementation could claim compliance without a determinate test for refutation. Phase D5 without Phase D2 produces tests whose targets are underspecified—the tests reference commitments whose content requires Phase D2's elaboration. Together, the two phases provide the complete prior-art record: a requirement set and a verification methodology jointly covering Paper 3's FAI governance claims.

2. The Sixty-Test Operational Library

Phase D5 expanded the D2.24 baseline of 20 tests to a library of 60 tests organized across six categories. The categories and their test ranges are as follows.

Tests 1–20 (D2.24 baseline). The foundational test set covering the core FAI governance dimensions: lifecycle governance (event initiation, shared substrate construction, dissolution), structural properties (aspect-unit exchange, layer bounding, instinct exclusion), exchange bounding (what crosses the inter-Self perimeter and what does not), conflict handling (preserve tier, orchestration tier, escalation tier), evolution feed routing (action-feedback path, DNA evolution path, four-locus accounting), perimeter integrity (home perimeter preservation during FAI), and path retraceability across the inter-Self boundary. These 20 tests constitute the minimum complete test suite for any FAI governance claim.

Tests 21–30 (D5.02, composition pair tests). Tests covering interactions between Paper 3 commitments—how the shared substrate construction interacts with conflict handling, how the three-tier conflict mechanism interacts with the evolution feed, how configurable dimensions interact with human authority requirements. These tests verify that the commitments hold not only in isolation but under their known interaction patterns.

Tests 31–40 (D5.03, anti-pattern detection tests). Tests designed to detect the named failure modes developed in Phase D3. Each test is keyed to a specific anti-pattern: opaque agent-to-agent

communication falsely presented as substrate-mediated, instinct-layer content crossing the inter-Self perimeter, conflict flattening under FAI merge, evolution feed bypass, governance authority delegated to the LLM rather than retained by humans. An implementation that passes the positive commitment tests but fails an anti-pattern detection test carries a latent governance deficiency the positive tests did not surface.

Tests 41–45 (D5.04, governance quality tests). Tests for governance quality properties that go beyond binary commitment satisfaction: consistency of governance configuration across multiple FAI events, non-specialist accessibility of governance inspection and override, path retraceability depth sufficient for after-action reconstruction, and authority articulation clarity at the inter-Self boundary. These tests differentiate minimally compliant implementations from robustly governed ones.

Tests 46–50 (D5.05, evolution learning tests). Tests verifying that FAI events produce properly routed evolution outputs—that action-feedback records enter each participating Self's home substrate, that DNA evolution absorption occurs under governance authorization, that evolution feed provenance carries through to home substrate annotations, and that instinct evolution is correctly excluded from the FAI evolution feed.

Tests 51–55 (D5.06, population-scope tests). Tests for Paper 3's Claim 6 territory: population-scale governance dynamics across many Selves. These tests verify that the architecture's commitments remain operative at population scope—that governance does not degrade as participating-Self count grows, that conflict handling scales without authority fragmentation, and that population-level evolution dynamics preserve each Self's home governance perimeter.

Tests 56–60 (D5.10, cross-paper tests). Tests verifying that Paper 3's commitments honor the inheritance constraints Paper 1 and Paper 2 established. These tests confirm that FAI governance inherits Paper 1's six architectural commitments within shared substrate scope, that the instinct/reasoning separation from Paper 2 is preserved across the inter-Self perimeter, and that the substrate-mediator commitment extends to the inter-Self scope without breaking the cell-scope and Self-scope properties the prior papers defend.

3. Ten Prior-Art Claims

Phase D5.14 consolidated ten prior-art claims that together establish Paper 3's governance commitments as operationally precise and verifiable:

1. FAI governance requirements are binary-testable at each commitment individually.
2. The 60-test library covers all six Paper 3 claims without overlap or gap.
3. Each test is derivable by non-specialist practitioners from Paper 3 and the Phase D2 elaboration.
4. Anti-pattern detection tests and positive commitment tests constitute distinct and non-redundant prior-art layers.

5. Governance quality tests establish verifiable criteria beyond binary compliance.
6. Evolution feed tests provide specific prior art against claims that FAI outputs enter home substrates without governance routing.
7. Population-scope tests establish that the architecture's verifiability claims extend to Claim 6's population-scale territory.
8. Cross-paper tests establish that Paper 3's governance inheritance from Papers 1 and 2 is verifiable, not merely asserted.
9. The test library as a whole constitutes a complete verification methodology for FAI governance—prior art against any claim that Paper 3's commitments are aspirational rather than testable.
10. Scenario-specific subset guidance (D5.09) demonstrates that the 60-test library is operationally deployable, not merely theoretical.

4. Combined Series D Scope Through Phase D5: 191 Notes

Phase D5's completion brings the combined Series D scope through note #650 to 191 notes.

Phase	Note Range	Count	Coverage
D0	#460–#465	6	Paper 3 claim anchors
D1	#466–#495	30	Foundational sub-commitments
D2	#496–#575	80	Governance requirements and operational variants
D3	#576–#605	30	Failure modes and anti-pattern formalizations
D4	#606–#635	30	Multi-commitment composition pairs
D5	#636–#650	15	Operational test library and verification methodology
Total	#460–#650	191	Complete prior-art foundation

These 191 notes constitute a complete operational prior-art foundation for Paper 3's FAI governance architecture. The four substantive phases address the four questions any governance architecture must answer: What must governance do? (D2, 80 notes.) What does governance failure look like? (D3, 30 notes.) How do governance commitments interact? (D4, 30 notes.) How is governance compliance verified? (D5, 15 notes.) The anchor phase (D0, 6 notes) and foundational phase (D1, 30 notes) provide the claim-level and commitment-level foundation on which the four substantive phases build. No substantive territory within Paper 3's governance claims is left without prior-art coverage at this point in the Series D program.

5. Transition to Phase D6: Boundary Cases as the Final Precision Layer

Phase D6 (notes #651–#665, approximately 15 notes) covers boundary cases—governance situations at the edges of Paper 3's commitments where the architecture must make explicit choices about which side of a boundary a scenario falls on.

Phase D6 is not a fifth substantive category parallel to D2–D5. It is a precision layer over the existing prior-art foundation. The four substantive phases address standard cases: FAI events that clearly fall within the architecture's scope, failure modes that clearly instantiate a named anti-pattern, commitment interactions that occur in typical governance configurations, test applications to recognizable implementations. Phase D6 addresses edge conditions where scope is not immediately obvious—the cases an adversarial reader would identify as potential grey areas.

Three families of boundary cases constitute Phase D6's coverage. The minimum edge: what is the simplest FAI event that still constitutes valid FAI governance? A single-aspect, one-time, floor-level event—one Self contributing one aspect, resolved without conflict, producing a minimal evolution feed—either satisfies the architecture's commitments or it does not. Establishing the minimum explicitly closes the argument that the architecture's commitments are only satisfied by complex, multi-Self, multi-aspect events. The maximum edge: what is the most comprehensive valid FAI governance? A full-Self contribution, many Selves, maximum governance depth—establishing that the architecture handles maximum complexity without commitment degradation closes the argument that governance breaks down at scale. The boundary definiteness cases: governance situations where the architecture must explicitly determine on which side of a named boundary a scenario falls—when does an exchange constitute a full FAI event versus an adjacent coordination pattern, when does conflict handling reach the escalation tier versus remaining in the orchestration tier, when does evolution feed absorption require explicit governance authorization.

Phase D6 forecloses the adversarial strategy of identifying grey areas as evidence that the architecture's scope is indeterminate. Each note in Phase D6 takes a specific boundary case and establishes, from Paper 3's commitments and the Phase D2–D5 elaboration, where it falls—not as a judgment call, but as a derivation from already-defended positions. The result is an architecture whose scope is determinate at all edges, not only at its center.

6. Conclusion

D5.15 closes Phase D5 as the fifteenth and final note in the phase. Phase D5 produced a 60-test operational library across six categories, ten prior-art claims, and verification methodology infrastructure that together constitute the verification counterpart to Phase D2's specification. Combined with Phases D0 through D4, the Series D program through Phase D5 totals 191 notes—a complete prior-art foundation covering Paper 3's FAI governance architecture at the levels of claims, sub-commitments, governance requirements, failure modes, commitment interactions, and verifiability. Phase D6 follows as the final precision layer, establishing boundary case determinations that close the grey-area adversarial gap and complete Series D's prior-art coverage of Paper 3.

Source paper

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